Parametrics

Section 10.3

3-26-26

A) Parametric Equations

Ex:

$x(t) = t^2 + 9$

$y(t) = 4t^5 - 6t^3 + 3$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{y'(t)}{x'(t)} = \frac{20t^4 - 18t^2}{2t}$$

$$= \boxed{10^2 - 9t}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d\,\frac{dy}{dx}}{dx}$$

$$= \boxed{\frac{30t^2 - 9}{2t}}$$

Ex. $x(t) = 5\sin t$ equation of tangent line

$y(t) = 3\cos t$

$x\left(\frac{2\pi}{3}\right) = \frac{5\sqrt{3}}{2}$ $y\left(\frac{2\pi}{3}\right) = -\frac{3}{2}$

$$\frac{dy}{dx} = \frac{-3\sin t}{5\cos t} = -\frac{3}{5}\tan t$$

$$\left.\frac{dy}{dx}\right|_{t=\frac{2\pi}{3}} = -\frac{3}{5}(-\sqrt{3})$$

$$y + \frac{3}{2} = \frac{3\sqrt{2}}{5}\left(x - \frac{5\sqrt{3}}{2}\right)$$

Find all points of Hor and Vert tangent lines.

HT: $\frac{dy}{dx} = 0$	VT
$y'(t) = 0$	$x'(t) = 0$
$2t - 8 = 0$	$6t - 6 = 0$
$t = 4$	$t = 1$
$(24, -7)$	$(-3, 2)$

cannot be the same

B) Arc Length

arclength formula: $\int_a^b \sqrt{1 + (f'(x))^2}\, dx$

or

$$\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Find arclength of

$x(t) = e^{5t+2}$ $0 \le t \le 4$

$y(t) = 3\sin t^2$

$$\int_0^4 \sqrt{\left(5e^{5t+2}\right)^2 + \left(6t\cos(t^2)\right)^2}\, dt$$